

Growth, Transformation and Transition

Business Economics, Climate Change and the Environment

Simon Quinn

Welcome to **Business Economics, Climate Change and the Environment!**

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This module aims to give a **general overview of the subject of economics and its relevance for climate change and the environment.**

Today's objectives

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Overview of the module

1. Outline of content
2. Lectures
3. Lecture slides
4. Textbook
5. Office hours
6. Online forum
7. Assessment: Exam
8. Assessment: Participation
9. Assessment: 'Academic Papers Explained'

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Module outline

LECTURE	DATE	TOPIC
1	9 September	Growth, Transformation and Transition
2	11 September	Introduction to Game Theory
3	16 September	Game Theory and Auctions
4	17 September	Game Theory and the Environment
5	23 September	Academic Papers Explained
6	25 September	Academic Papers Explained
7	30 September	Consumer Preferences
8	2 October	Institutions, Power and Inequality
9	7 October	Supply and Demand
10	9 October	Market Successes and Failures

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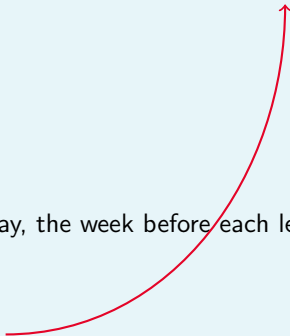
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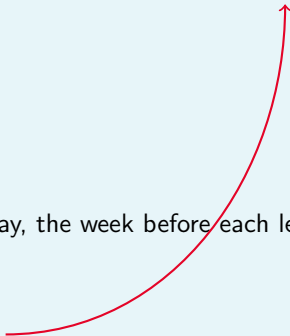
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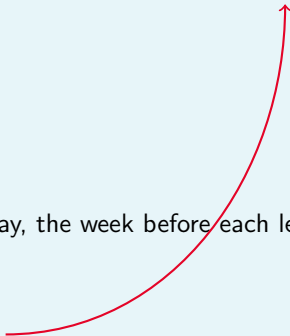
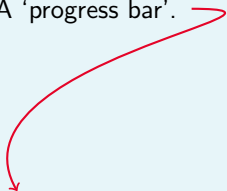
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I will not 'teach the textbooks' as such; some lectures will be close to the textbook, and some will be less close. The textbooks are an excellent **support** for our lectures.

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Office hours

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(You are welcome to chat to me during the lecture break too, of course!)

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I will follow the general discussion on the forum, but the forum is **not** the right place for asking me questions – I cannot guarantee that I will see it. (Instead, please use my office hours.)

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The exam will comprise a series of short-answer questions; these questions may require you to draw appropriate diagrams. All of the lectures are examinable, as well as the assigned readings. Note that you may choose to refer, in your exam answers, to content from the 'academic papers explained'; this may strengthen your answer by providing further real-world illustrations of the principles covered in the lecture slides.

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Note: There is a **huge** difference between **looking at** a diagram and being able to **replicate** it!

Assessment: Examination

The exam will last for two hours, and will be conducted in person. It will be conducted using WiseFlow for written answers; diagrams will be hand-drawn in booklets provided. There will be three parts to the exam:

Part A will relate to Lectures 2, 3 and 4 ('Introduction to Game Theory', 'Game Theory and Auctions' and 'Game Theory and the Environment'). Part A will have two questions; you will need to answer one of these.

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Part C will relate to Lectures 1, 9 and 10 ('Growth, Transformation and Transition', 'Supply and Demand' and 'Market Successes and Failures'). Part C will have two questions; you will need to answer one of these.

Exam details: Example exam

I will upload an example exam to Insendi ('Files' tab) in due course.

This will provide an illustration of the style of questions for the actual exam. It will also provide a summary of what an excellent answer might look like.

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(Please note that the example exam is **just an example** – it is designed to help you to prepare for the actual exam, but it is not intended to show you what is (or is not) on the actual exam.)

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This is measured by participation in the online forum: you will receive the full 5% if you make (at least) one substantive and thoughtful posting (for example, a thoughtful posting comprising a short paragraph on some issue discussed in lectures or readings). (You are welcome to respond to other students' postings on the forum; of course, please make sure that the tone of all postings is polite and constructive throughout.)

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Each group will be required to give a presentation about one academic paper in economics.
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All presentations will be given in the lecture, in front of the class; presentations will take place in Lecture 5 (23 September) and Lecture 6 (25 September).

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You are expected to attend in person for your group's presentation, **and** for the others groups' presentations.

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Each group must have slides to accompany the presentation; these can be in Powerpoint or PDF format. Please upload these to Insendi by 5pm on 22 September.

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- (v). Whether the presentation meets the 15-minute limit (marks to be deducted if the presentation is shorter than 14 minutes or longer than 16 minutes).

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Note: You are **not** required to cover formal models, nor to explain empirical techniques in detail. The emphasis should be on the general approach of the paper, and the results.

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Outline

1. GDP, growth and institutions over time
2. The climate crisis over time
3. Growth and structural transformation
4. The climate crisis as a policy challenge

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GDP and economic growth

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Definition: Economic growth

Increases in real GDP over time.

Some thought experiments about GDP

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5. The UK Government legalises smoking for children from age 10.

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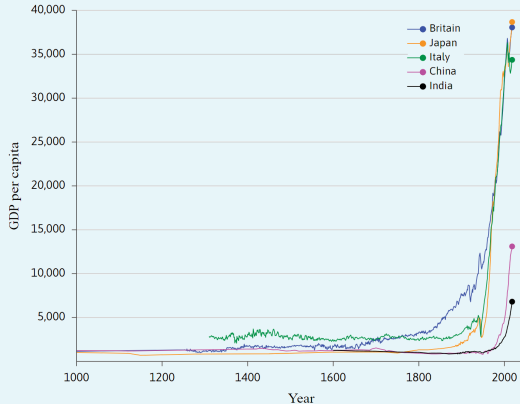
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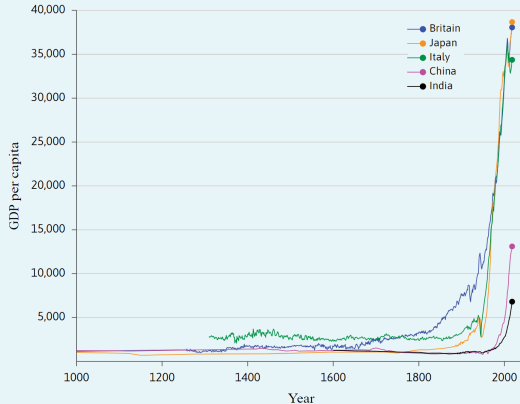
Is GDP a good **measure** of economic progress? Is GDP a good **objective** for government policy? Can you suggest some **better objectives**?

'The hockey stick'



This graph shows **real GDP per capita** over **time**.

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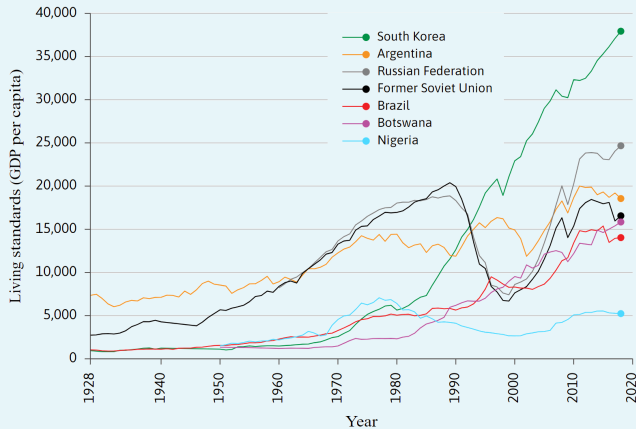
Questions...

Which countries took off first?
Why do you think they did so?

'200 Countries, 200 Years, 4 Minutes' (Rosling, 2011)

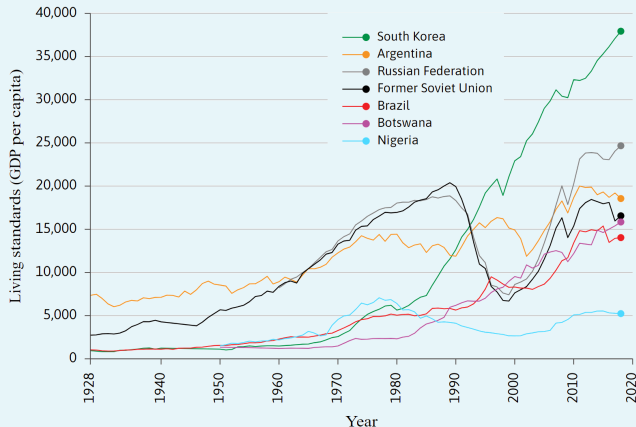
<https://www.youtube.com/watch?v=jbkSRLYSojo>

Varieties of capitalism



This graph shows real GDP per capita among a set of 'latecomers to the capitalism revolution'.

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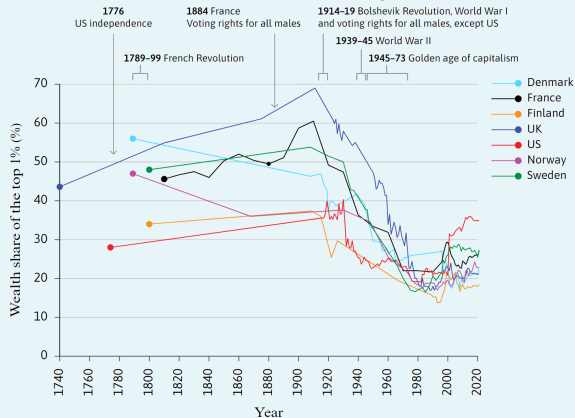


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Questions. . .

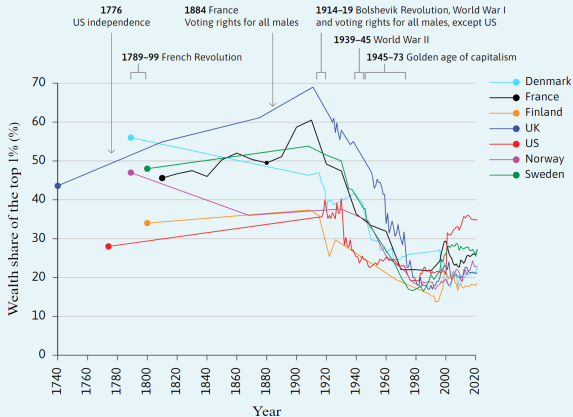
What do you see as the key trends here?

Capitalism, inequality and democracy



This graph shows the share of total wealth held by the top 1% (for all countries for which long-run data is available).

Capitalism, inequality and democracy



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Questions...

What do you see as the key trends here? How do these trends relate to the arrival of democracy?

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So the data seems to show that...

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Summary so far...

So the data seems to show that...

- ▶ Real GDP per capita is a useful measure (if very imperfect) for economic progress.
- ▶ There has been very rapid economic growth over the past century (or so).
- ▶ This growth has helped to lift large numbers of people out of severe poverty.

Summary so far...

So the data seems to show that...

- ▶ Real GDP per capita is a useful measure (if very imperfect) for economic progress.
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Time for some more hockey sticks...

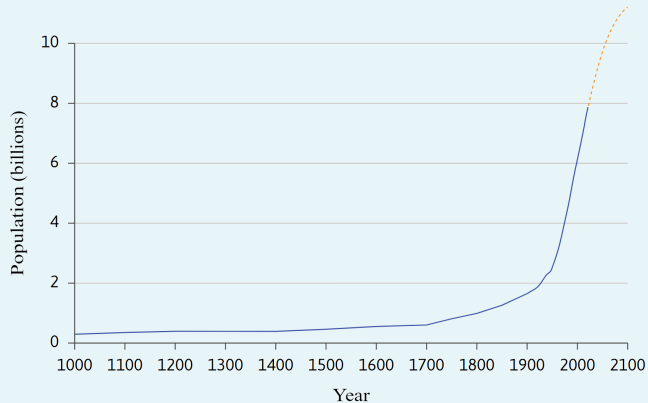
Outline

1. GDP, growth and institutions over time
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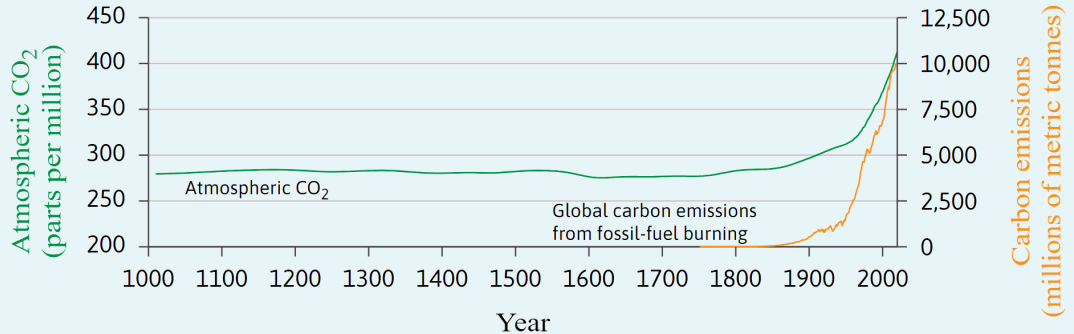
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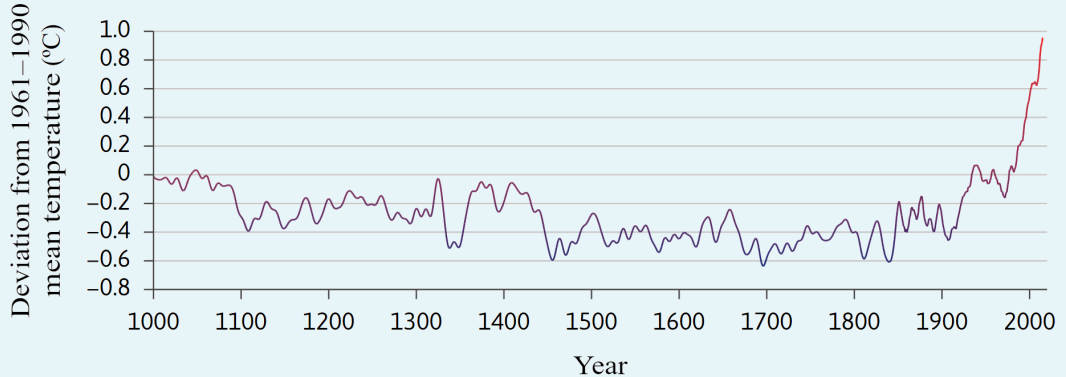
The global population hockey stick



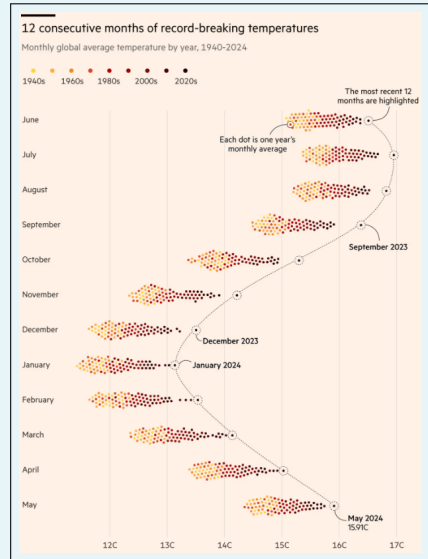
Carbon dioxide and global carbon emissions



Climate change over a millennium



Climate change over 85 years...

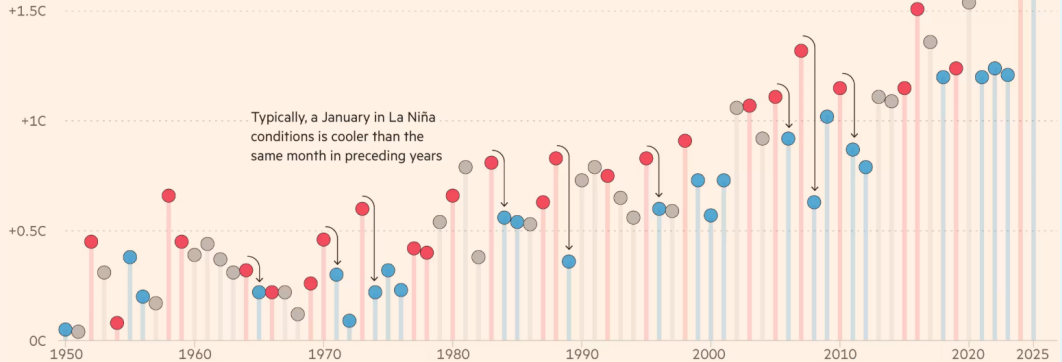


*'The dangerous effects of rising sea temperatures',
Financial Times, 24 July 2024*

January 2025 was the warmest on record despite an emerging La Niña

Global average January temperature per year compared with 1850-1900 average

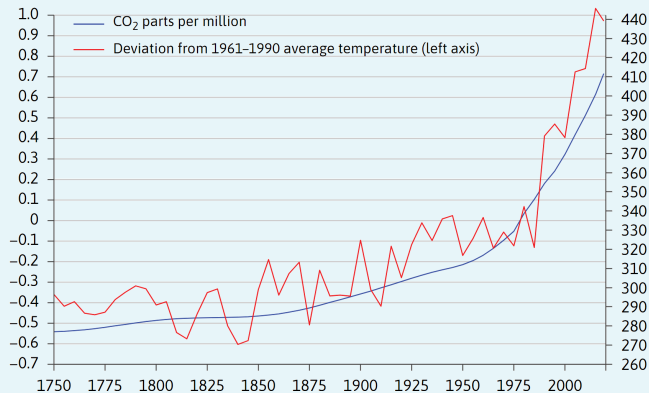
Conditions: ● La Niña ● Neutral ● El Niño



Source: ERA5, C3S/ECMWF
© FT

'Hottest January on record shocks scientists', Financial Times, 6 February 2025

Carbon dioxide and global warming



Human-caused climate change

Widespread and rapid changes in the atmosphere, ocean, cryosphere and biosphere have occurred.

Intergovernmental Panel on Climate Change, 2023

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Intergovernmental Panel on Climate Change, 2023

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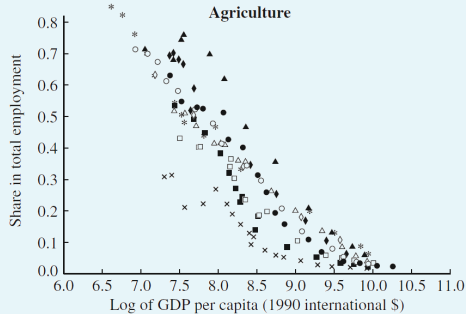
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Structural transformation and economic growth

So far, we have talked about **economic growth**, and we have seen striking evidence of **human-caused climate change**.

But how does economic growth actually happen? Can more growth ultimately be good for the environment?

Structural transformation and economic growth

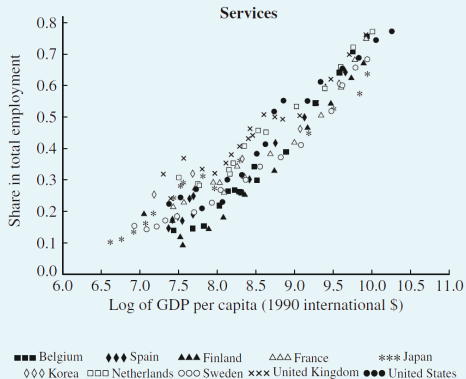


This graph compares the share of employment in **agriculture** with GDP per capita (on a log scale; more on this shortly).

Note that this combines evidence **across countries** and **across time**.

(Herrendorf, Rogerson and Valentinyi, 2014)

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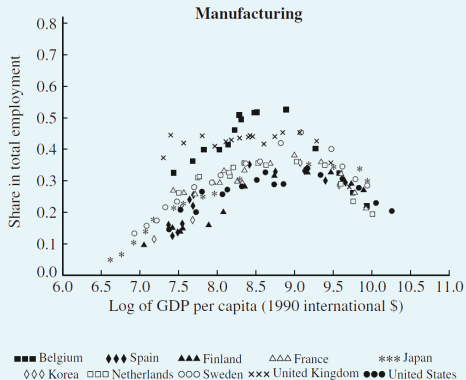
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What about employment in **services**?

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What about employment in **services**?

But now – what about **manufacturing**? (And why might this matter for the environment?)

An 'Environmental Kuznets Curve'?

The environmental Kuznets curve (EKC) is a hypothesized relationship between various indicators of environmental degradation and income per capita.

David Stern, '*The environmental Kuznets curve after 25 years*', 2017

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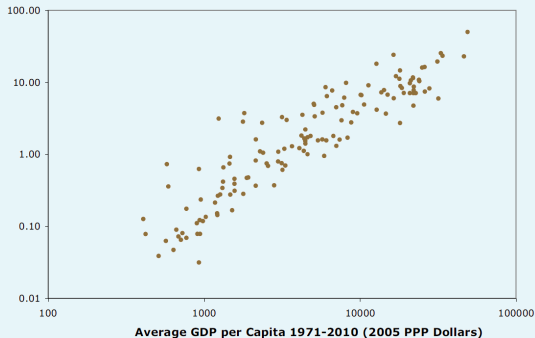
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An 'Environmental Kuznets Curve'?

Average per capita CO₂ emissions, 1971-2010 (tonnes)



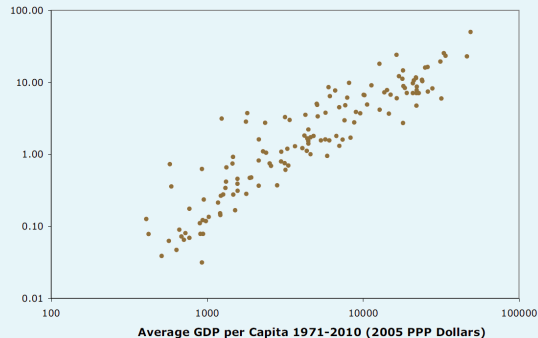
We now look at several measures of environmental damage. Each point refers to a country.

(Note the use of **log scales** here.)

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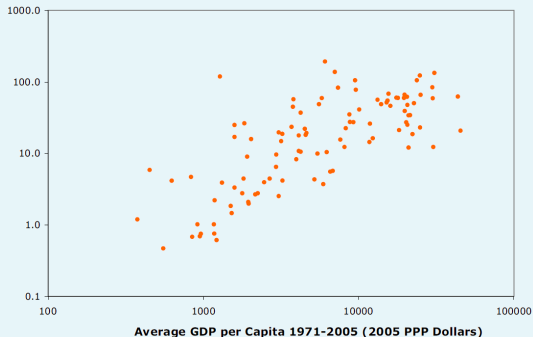
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Let's start with **carbon dioxide emissions**. Do you see an Environmental Kuznets Curve?

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An 'Environmental Kuznets Curve'?

Average per capita sulfur emissions, 1971-2005 (kg)



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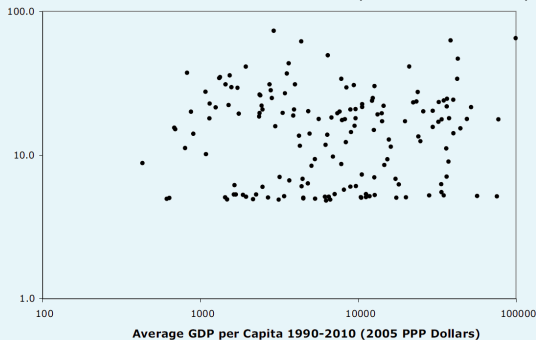
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Average PM2.5 concentration, 1999-2010 (micrograms per m³)



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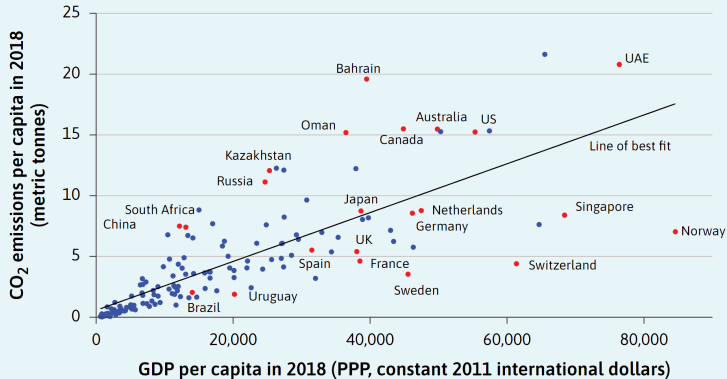
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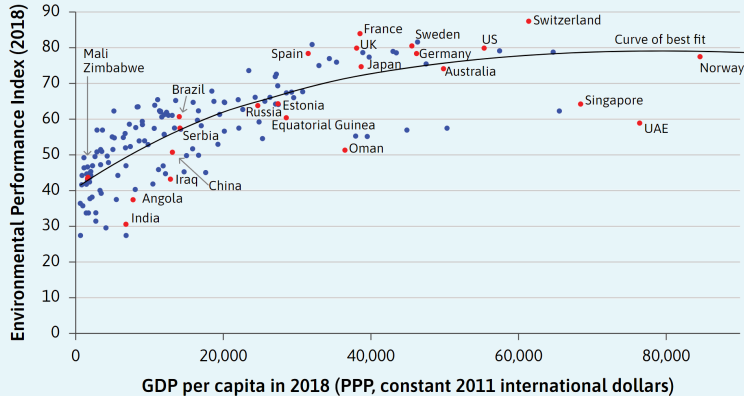
What about for **sulfur emissions**?

And **particulate matter**?

‘Carbon dioxide emissions are higher in richer countries...’



‘... but so is the quality of their local environment.’



‘The future ain’t what it used to be...’

We need to be careful in our analysis. The graphs we have seen look at **past economic growth** – but policy might encourage **future economic growth** to be different.

'The future ain't what it used to be...'

We need to be careful in our analysis. The graphs we have seen look at **past economic growth** – but policy might encourage **future economic growth** to be different.

Green growth means fostering economic growth and development while ensuring that natural assets continue to provide the resources and environmental services on which our well-being relies.

Organisation for Economic Cooperation and Development, 2011

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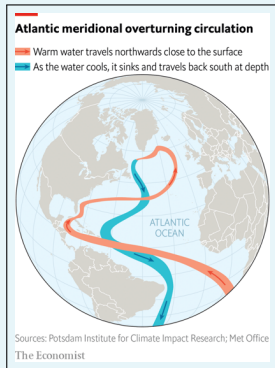
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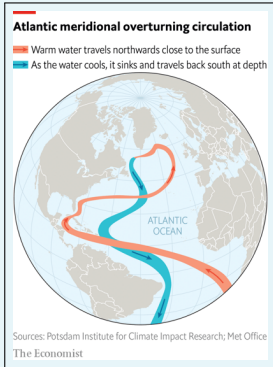
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However, **future environmental dynamics** might also be different...

The Atlantic Meridional Overturning Circulation

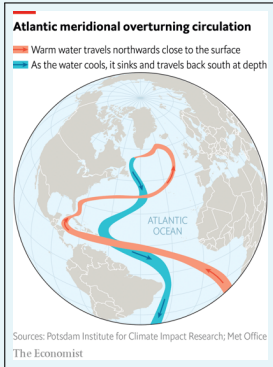


The Atlantic Meridional Overturning Circulation



“[AMOC’s] **tipping-point nature is not open to question**. Theory, modelling and reconstructions of prehistoric climate all support the idea that AMOC is “bistable”. **Rather than just gradually getting stronger or weaker, it can go suddenly from “on” to “off” if pushed too far**... On July 25th a paper published in *Nature Communications* suggested that the change of state could come about by the middle of this century...

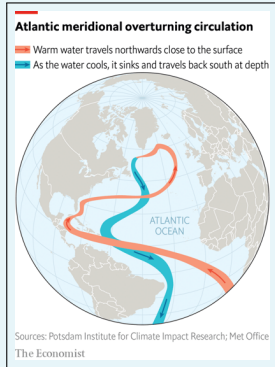
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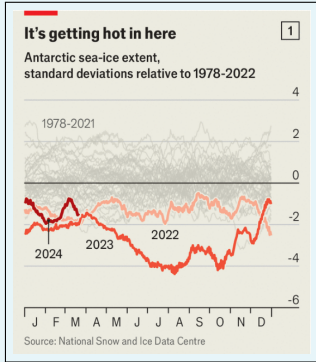


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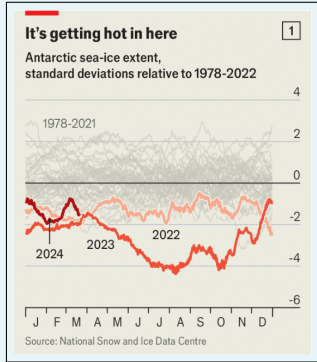
‘What is AMOC, the heat-distributing Atlantic current?’, *The Economist*, 2 August 2023

Antarctic sea ice: A tipping point?



*'Antarctica, Earth's largest refrigerator, is defrosting',
The Economist, 27 March 2024*

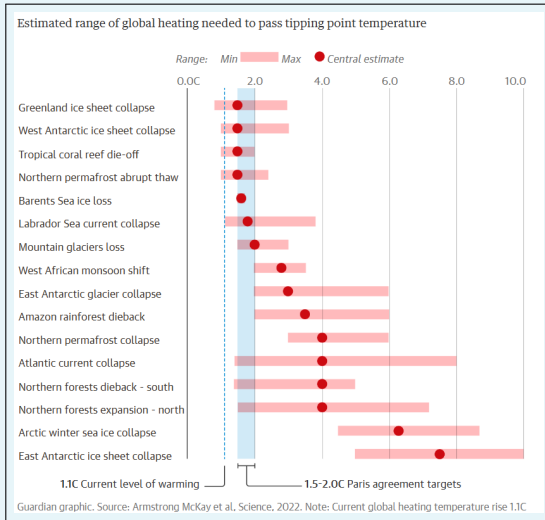
Antarctic sea ice: A tipping point?



“In June, the British Antarctic Society said it had observed a ‘new and worrying way’ that large ice sheets can melt, as relatively warm seawater laps at the underside of ground-based ice and accelerates its movement into the ocean. It warned that this was a **possible new tipping point** in Antarctic ice sheet melting that had not been taken into account in models before.”

*‘Antarctica, Earth’s largest refrigerator, is defrosting’,
The Economist, 27 March 2024*

*‘The dangerous effects of rising sea temperatures’,
Financial Times, 24 July 2024*



'World on brink of five 'disastrous' climate tipping points, study finds', The Guardian, 8 September 2022

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5. **Conflicts of interest**

The impacts of climate change differ among people according to their economic circumstances, both across the globe and within countries. Future generations will experience the effects of today's emissions, but also the actions we take to reduce them.

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Module outline

LECTURE	DATE	TOPIC
1	9 September	Growth, Transformation and Transition
2	11 September	Introduction to Game Theory
3	16 September	Game Theory and Auctions
4	17 September	Game Theory and the Environment
5	23 September	Academic Papers Explained
6	25 September	Academic Papers Explained
7	30 September	Consumer Preferences
8	2 October	Institutions, Power and Inequality
9	7 October	Supply and Demand
10	9 October	Market Successes and Failures

Growth, Transformation and Transition

Business Economics, Climate Change and the Environment

Simon Quinn